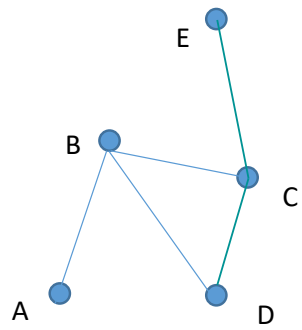
Lesson 6 - The Königsberg Bridge Problem

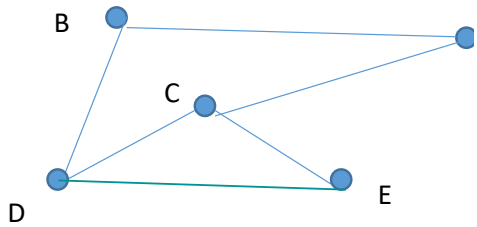
6.1 Walk



Question 71

Level of difficulty: E

Decide whether it is an open or closed walk and state the length in each case: a) ABCE b) BDCB c) BDCE



Question 72

Level of difficulty: E

Decide whether it is an open or closed walk and specify the length in each case:
a) DBAC b) CEDCABDC c) DBACD

Solution question 71

Solution hint: Count the number of edges and check if the walk returns to the initial node.

Solution: a) open walk of length 3, b) closed walk of length 3, c) open walk of length 3

Solution question 72

Solution hint: Count the number of edges and check if the path returns to the initial node.

Solution: a) open walk of length 3, b) closed walk of length 7, c) closed walk of length 4

Question 73

Level of difficulty: M

The following walk belongs to a graph: DCAD closed walk of length 3, ABDCA closed walk of length 4, CADB open walk of length 3. Draw a possible graph for this purpose.

Question 74

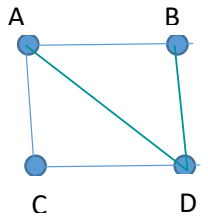
Level of difficulty: M

The following walk belong to a graph: BACD open walk of length 3, ACDA closed walk of length 3, CADCAB open walk of length 5. Draw a possible graph for this purpose.

Solution question 73

Solution hint: Draw the nodes first.

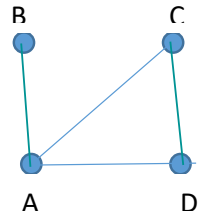
Solution: Several solutions are possible.



Solution question 74

Solution hint: Draw the nodes first.

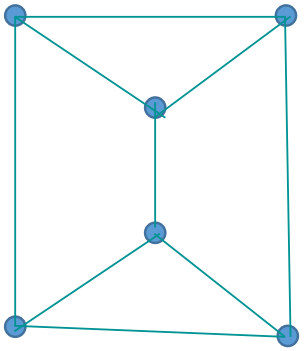
Solution: Several solutions are possible.



Question 75

Level of difficulty: D

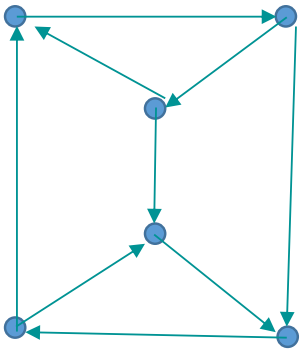
Show that the following graph leads to a strongly connected digraph by giving all edges suitable directions.



Solution question 75

Solution hint: Look up what is meant by a strongly connected digraph.

Solution:

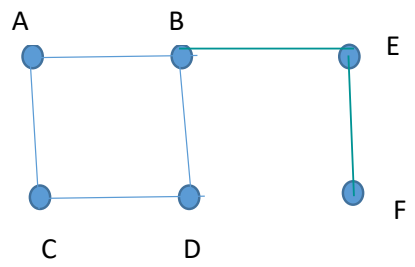


6.2 Eulerian graph

Question 76

Level of difficulty: E

Add an edge to the following graph so that it becomes Eulerian.



Question 77

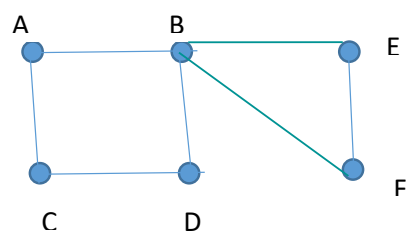
Level of difficulty: E

Is the Nicholas house an Eulerian graph? Justify your answer.

Solution question 76

Solution hint: If the two nodes with odd degree are connected by an additional edge, it is possible to return to the starting point and to draw one or more Eulerian cycles.

Solution: So that all node degrees are straight, an additional edge must be inserted between nodes B and F.



Solution question 77

Solution hint: All node degrees must be even.

Short solution: No

Detailed solution: No, because two nodes have an odd degree.

Question 78

Level of difficulty: M

Is the wheel graph W_5 an Eulerian path, an Eulerian cycle, or neither?

Question 79

Level of difficulty: M

Is the complete graph K_5 an Eulerian cycle? Justify your answer.

Solution question 78

Solution hint: Draw W_5. Which conditions must be met for an Eulerian path or an Eulerian cycle to exist?

Short solution: No

Detailed solution: The wheel graph W_5 is neither an Eulerian path nor an Eulerian cycle. It has four nodes with odd degree 3. Prerequisite for an Eulerian path are max. two nodes with odd degree. The prerequisite for an Eulerian cycle is only nodes with even degree.

Solution question 79

Solution hint: Draw the complete graph. Which conditions must be met for an Eulerian cycle to exist?

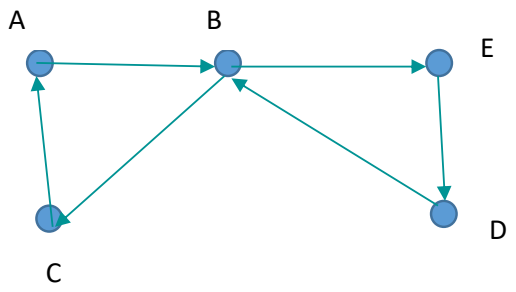
Short solution: Yes

Detailed solution: Yes, since it has only nodes with even node degree. All complete graphs with n odd are Eulerian.

Question 80

Level of difficulty: D

Is the following digraph Eulerian? Give reasons for your answer.



Solution question 80

Solution hint: A digraph is Eulerian if and only if it is strongly connected and for each node it holds that the input degree is equal to the output degree.

Short solution: Yes

Detailed solution: Yes, the digraph is strongly connected and for each node the input and output degrees match.

6.3 Hierholzer's algorithm

Question 81

Level of difficulty: E

What is Hierholzer's algorithm used for? Explain the procedure of this algorithm.

Question 82

Level of difficulty: E

What else is Hierholzer's algorithm called? Why is it called that?

Solution question 81

Solution hint: Eulerian cycle

Solution: Hierholzer's algorithm is used to determine Eulerian cycles in an undirected graph. Prerequisites: Connected graph that has only nodes with even degree.

1. Choose a node and construct from it a subcycle K that does not pass through any edge twice.
2. If K is an Eulerian cycle, then break off. Otherwise, continue to step 3.
3. Neglect all edges of K.
4. At the first vertex of K, let a second subcycle K_2 arise which contains no edge of K and no edge of the original graph twice.
5. Insert the second circle K_2 into K.
6. Continue with step 2.

Solution question 82

Solution hint: Familiarize yourself with the algorithm's procedure.

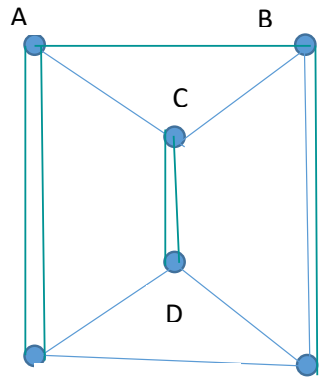
Short solution: Onion skin algorithm

Detailed solution: Onion skin algorithm. The constructed closed walk resembles nested onion skins.

Question 83

Level of difficulty: M

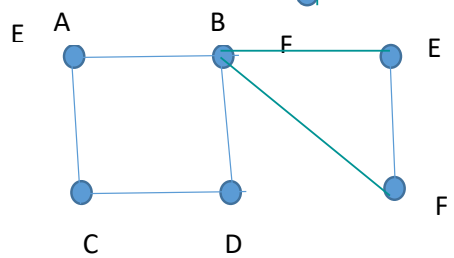
First check whether the given undirected graph is Eulerian. If so, then run Hierholzer's algorithm and determine an Eulerian cycle.



Question 84

Level of difficulty: M

First check whether the given graph Eulerian.
If so, then run Hierholzer's algorithm and determine an Eulerian cycle.



Solution question 83

Solution hint: Which conditions must be given for an Eulerian cycle? Look at the procedure for Hierholzer's algorithm.

Short solution: Yes, AEDFBACBFEA

Detailed solution: The given graph is Eulerian since it contains only nodes with even node degree. Hierholzer's algorithm yields an Eulerian cycle AEDFBACBFEA. Multiple solutions are possible.

Solution question 84

Solution hint: Which conditions must be given for an Eulerian cycle? Look at the procedure for Hierholzer's algorithm.

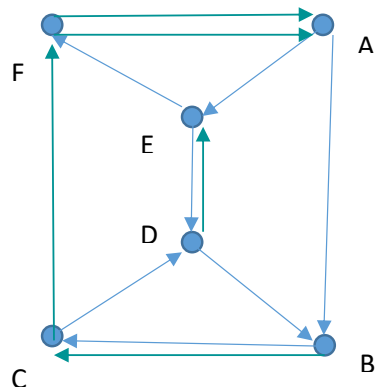
Short solution: Yes, ABEFBDCA

Detailed solution: The given graph is Eulerian since it contains only nodes with even node degree. Hierholzer's algorithm yields an Eulerian cycle ABEFBDCA. Several solutions are possible.

Question 85

Level of difficulty: D

First check whether the given directed graph is Eulerian. If so, then run Hierholzer's algorithm and determine an Eulerian cycle.



Solution question 85

Solution hint: Which conditions must be given for an Eulerian cycle? Look at the procedure for Hierholzer's algorithm.

Short solution: Yes, ABCDBCFAEDEFA

Detailed solution: The given graph is Eulerian since it is strongly connected and at each node the input degree coincides with the output degree. Hierholzer's algorithm yields an Eulerian cycle ABCDBCFAEDEFA.

6.4 The Postman Problem

Question 86

Level of difficulty: E

Explain the postman problem. Give two concrete examples from practice.

Question 87

Level of difficulty: E

Explain in simplified terms the approach to the postman problem.

Solution question 86

Solution hint: Put yourself in the shoes of a postman.

Short solution: Walk all streets with minimal effort and return to starting point; examples include delivery services and street cleaning.

Detailed solution: A postman has to divide accurately his working time. His delivery area comprises several streets through which he must pass at least once. Detours or return routes, if they cannot be avoided, should be as short as possible. The streets are considered edges and the intersections and junctions are considered nodes. The graph is weighted, since the length of the streets plays a role. We are looking for a closed set of edges that contains all streets of the delivery area and has a minimum total weight. Similar problems are generally found in collection and delivery services, such as garbage collection, street cleaning, or factory traffic, but also in city tours, etc.

Solution question 87

Solution hint: Find the shortest walk between every two nodes with odd degree.

Solution: The shortest walk between every two nodes with odd degree, i.e., every pair, is to be found by applying Dijkstra's algorithm. The edges of this walk are to be doubled. The newly created graph is Eulerian, so that an Eulerian cycle is to be found in it using Hierholzer's algorithm.

Question 88

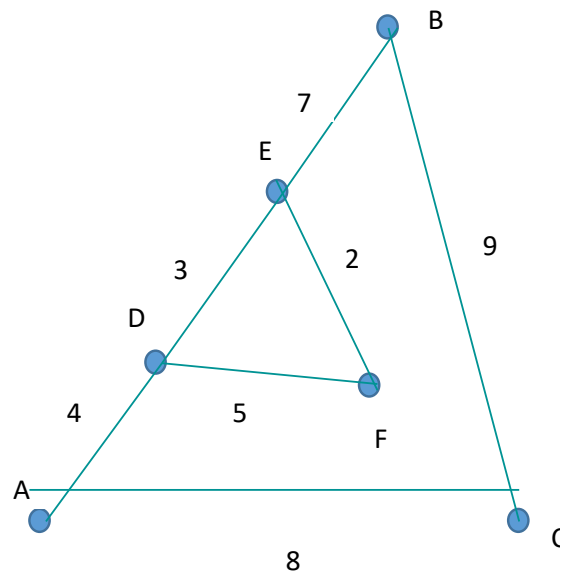
Level of difficulty: M

Explain the solution of the postman problem using graph theory.

Question 89

Level of difficulty: M

For the following graph, give the optimal solution to the postman problem.



Solution question 88

Solution hint: The shortest route corresponds to an Eulerian cycle.

Short solution: An Eulerian cycle is optimal. If the graph is not Eulerian, it must be made Eulerian by adding edges.

Detailed solution: In Eulerian graphs, the shortest route corresponds to an Eulerian cycle. By definition, an Eulerian cycle contains each edge exactly once. However, the problem requires that each edge be contained at least once. Therefore, any solution that contains each edge exactly once is optimal in every case. If the graph is not Eulerian, the problem is solved by making the graph Eulerian by adding optimal edges. For the postman, duplicated edges mean that these roads must be walked twice. If the graph is Eulerian, we proceed as above and we look for an Eulerian cycle as the optimal solution. For a connected graph that is not Eulerian, there are an even number of nodes that have odd degree. If each two nodes of odd degree are connected by an additional shortest walk doubling all edges in this walk, the formerly odd node degrees become even. Nodes with already even node degree within the walk remain even by the doubling. The shortest walk between two nodes of odd degree can be determined using Dijkstra's algorithm.

Solution question 89

Solution hint: Double the edges with odd node degree.

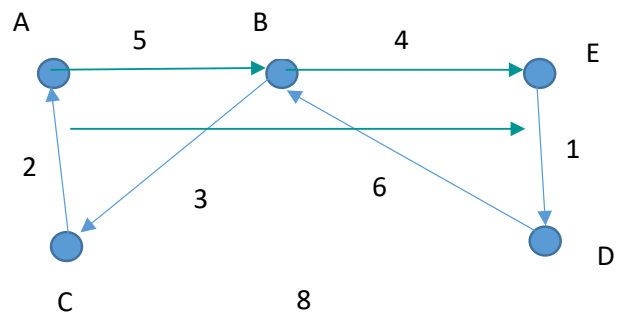
Short solution: Double the edge between D and E; Eulerian cycle: ADEFDEBCA

Detailed solution: D and E are nodes with odd degree. The shortest edge from D to E is the direct one. Thus, this edge is to be doubled. Hierholzer's algorithm yields the following Eulerian cycle: ADEFDEBCA. This is a minimal postman path. Note: The Eulerian cycle need not be unique.

Question 90

Level of difficulty: D

For the following digraph, enter a minimum cost solution for the postman problem.



Solution question 90

Solution hint: Input and output degree must match.

Short solution: Additional arrows are (D, B) and (B, C); Eulerian digraph ABCDBEDBCA.

Detailed solution: At nodes C and D, the input and output degrees do not match. At C another arrow should arrive, at D another arrow should leave. So the shortest directed walk from D to C is searched. This leads over B with length 9. So the additional arrows are (D, B) and (B, C). An Eulerian cycle in the modified Eulerian digraph is ABCDBEDBCA.